

Cryptanalysis of Vigenère Cipher using Kasiski Examination and Frequency Analysis

Your program must:

1. Preprocess the ciphertext (Given as per group number)
2. Estimate the key Length using Kasiski's test
3. Divide the ciphertext into groups according to the estimated key length.
4. Perform frequency analysis on each group.
5. Determine the probable key.
6. Decrypt the ciphertext using the recovered key.
7. Display:
 - o estimated key length
 - o frequency table for each group
 - o recovered key
 - o recovered plaintext
8. Verify the result by re-encrypting the recovered plaintext and checking that it produces the original ciphertext.

Your program must have these user-defined functions

Function	Purpose
clean_ciphertext()	Remove spaces/special characters and normalize the ciphertext
find_repeated_patterns()	Identify repeated sequences in the ciphertext
calculate_distances()	Find distances between repeated occurrences
find_factors()	Find factors of the distances obtained from repeated patterns
kasiski_analysis()	Use repeated patterns/distances to suggest candidate key lengths
calculate_ic()	Calculate Index of Coincidence
split_into_groups()	Divide ciphertext according to a candidate key length
frequency_analysis()	Calculate A–Z frequency for each group
find_shift()	Estimate the Caesar shift for a group
find_key()	Combine shifts to obtain the probable Vigenère key
vigenere_decrypt()	Decrypt ciphertext using the recovered key
vigenere_encrypt()	Re-encrypt plaintext for verification
verify()	Check whether re-encryption produces the original ciphertext

Ciphertext 1 (Odd Group No.)

DAZFI SFSPA VQLSN PXYSZ WXALC DAFGQ UISMT PHZGA
MKTTF TCCFX
KFCRG GLPFE TZMMM ZOZDE ADWVZ WMWKV GQSOH QSVHP
WFKLS LEASE
PWHMJ EGKPU RVSXJ XVBWV POSDE TEQTX OBZIK WCXLW
NUOVJ MJCLL
OEOFA ZENVM JILOW ZEKAZ EJAQD ILSWW ESGUG KTZGQ
ZVRMN WTQSE
OTKTK PBSTA MQVER MJEGL JQRTL GFJYG SPTZP GTACM
OECBX SESCO
YGUFP KVILL TWDKS ZODFW FWEAA PQTFS TQIRG MPMEL
RYELH QSVWB
AWMOS DELHM UZGPG YEKZU KWTAM ZJMLS EVJQT GLAWV
OVVXH KWQIL
IEUYS ZWXAH HUSZO GMUZQ CIMVZ UVWIF JJHPW VXFSE
TZEDF

Ciphertext-2 (Even Group Numbers)

QRBAI UWYOK ILBRZ XTUWL EGXSN VDXWR XMHXY FCGMW
WWSME LSXUZ
MKMFS BNZIF YEIEG RFZRX WKUFA XQEDX DTTHY NTBRJ
LHTAI KOCZX
QHBND ZIGZG PXARJ EDYSJ NUMKI FLBTN HWISW NVLFM
EGXAI AAWSL
FMHXR SGRIG HEQTU MLGLV BRSIL AEZSG XCMHT OWHFM
LWMRK HPRFB
ELWGF RUGPB HNBEM KBNVW HHUEA KILBN BMLHK XUGML
YQKHP RFBEL
EJYNV WSIJB GAXGO TPMXR TXFKI WUALB RGWIE GHWHG
AMEWW LTAEL
NUMRE UWTBL SDPRL YVRET LEEDF ROBEQ UXTHX ZYOZB
XLKAC KSOHN
VWXKS MAEPH IYQMM FSECH RFYPB BSQTX TPIWH GPXQD
FWTAI KNNBX

SIYKE TXTLV BTMQA LAGHG OTPMX RTXTH XSFYG WMVKH
LOIVU ALMLD
LTSYV WYNVW MQVXP XRVYA BLXDL XSMLW SUIOI IMELI
SOYEB HPHNR
WTVUI AKEYG WIETG WWBVM VDUMA EPAUA KXWHK MAUPA
MUKHQ PWKCX
EFXGW WSDDE OMLWL NKMWD FWTAM FAFEA MFZBN WIHYA
LXRWK MAMIK
GNGHJ UAZHM HGUAL YSULA ELYHJ BZMSI LAILH WWYIK
EWAHN PMLBN
NBVPJ XLBEF WRWGX KWIRH XWWGQ HRRXW IOMFY CZHZL
VXNVI OYZCM
YDDEY IPWXT MMSHS VHHXZ YEWNV OAOEL SMLSW KXXFX
STRVI HZLEF
JXDAS FIE